TrafficSense — Multi-Agent Smart City Traffic Management System

Capstone Project | School of Computer Science & Engineering | VIT University

■ Student Information

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Register No.	23BCE8015
Programme	B.Tech — Computer Science and Engineering
University	VIT-AP University, Amaravati
Project Title	TrafficSense: A Multi-Agent Smart City Traffic Management System
Role	System Architect & Multi-Agent Systems Lead — responsible for overall project design, LLM-based agent cooperation framework, and CityFlow traffic simulation
Academic Year	FALL SEM 2026-27
Faculty Mentor	Dr. Mohd Irfan, Assistant Professor Sr. Grade-1, SCOPE, VIT-AP University

■ Contribution Overview

My primary responsibility in the TrafficSense project was designing the overall system architecture and implementing the Multi-Agent coordination layer. This is the core intelligence of the system — the component that makes traffic lights 'talk' to each other and make cooperative decisions in real time, rather than operating in isolation on fixed timers.

■ Detailed Technical Contributions

1. Overall System Architecture Design

- Designed the three-layer architecture: Perception → Analytics → Multi-Agent Control.
- Defined inter-module communication contracts and data schemas (JSON-based state objects).
- Established the project directory structure, module boundaries, and integration points.
- Created the *src/config.py* global configuration file used by all modules.

2. Multi-Agent System (MAS) Framework

• Implemented the agent base class in *src/agents/base_agent.py* defining the observe → reason → act loop.

• Built the intersection agent (*src/agents/intersection_agent.py*) — each intersection is an autonomous agent that holds its own state and makes local decisions.

• Developed the LLM-based reasoning engine: agents query a local LLM (via Ollama) using structured prompts that include real-time congestion data, queue lengths, and neighboring agent states.

• Implemented the cooperative communication protocol where agents broadcast their state to direct neighbors and negotiate green-light priorities.

3. CityFlow Traffic Simulation Integration

• Integrated CityFlow (a high-performance traffic simulator) as the simulation environment.

• Built the simulation runner in *src/simulation/cityflow_runner.py*.

• Designed the 4-intersection road network topology with realistic Indian traffic flow parameters.

• Implemented the 360-step simulation loop with sub-20-minute execution guarantee.

• Wrote the state extractor that converts raw CityFlow output into structured agent observations.

4. Emergency Vehicle Priority Logic

• Designed and implemented the green corridor algorithm: when the perception layer detects an ambulance, the multi-agent system cascades green signals along the predicted route.

• Implemented priority preemption — emergency vehicles override all other scheduling decisions.

■ Deliverables & Files

File / Module	Description	Status
src/config.py	Global project configuration	✔ Complete
src/agents/base_agent.py	Abstract agent base class	✔ Complete
src/agents/intersection_agent.py	LLM-driven intersection controller	✔ Complete
src/agents/coordinator.py	Multi-agent coordinator and message broker	✔ Complete
src/simulation/cityflow_runner.py	CityFlow simulation harness (360 steps)	✔ Complete
src/simulation/state_extractor.py	Converts simulation state to agent observations	✔ Complete
data/cityflow/	Road network JSON configs for 4 intersections	✔ Complete

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■ Technologies & Tools Used

Language	Python 3.12
Simulation	CityFlow (C++ traffic simulator with Python API)
LLM Engine	Ollama (local LLM inference) + structured JSON prompting
Agent Logic	Custom rule-based + LLM hybrid reasoning
Config	YAML / JSON
Version Control	Git + GitHub

■ Declaration

I, Achalla Akul Lakhan (23BCE8015), hereby declare that the contributions described in this report are my individual work carried out as part of the TrafficSense Capstone Project. All implementations listed above were personally designed, coded, tested, and integrated by me.

Signature: _____ Date: _____